

Demonstrating Sustainflatable: Harvesting, Storing and Utilizing Ambient Energy for Pneumatic Morphing Interfaces

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ABSTRACT

While the majority of pneumatic interfaces are powered and controlled by traditional electric pumps and valves, alternative sustainable energy harnessing technology has been attracting attention. This paper presents a novel solution to this challenge with the development of the Sustainflatable system, a self-sustaining pneumatic system that can harvest renewable energy sources such as wind, waterflow, moisture, and sunlight, convert the energy into compressed air, and store it for later use in a programmable and intelligent way. The system is completely electronic-free, incorporating customized energy harvesting pumps, storage units with variable volume-pressure characteristics, and tailored valves that operate autonomously. Additionally, the paper provides a design tool to guide the development of the system and includes several environmental applications to showcase its capabilities.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**.

KEYWORDS

pneumatic interface, energy harvesting, shape-changing interface

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1 INTRODUCTION

Renewable energy sources have been utilized by humans since ancient times, as evidenced by the use of windmills [8] and water-wheels [34]. However, with the rise of the industrial revolution, the

majority of our energy now comes from non-renewable fossil fuels such as oil, natural gas, and coal [10]. This dependence on non-renewable energy sources has resulted in environmental damage, energy crises, and climate change [5, 28].

To mitigate these problems, efforts have been made to increase the proportion of clean and renewable energy sources such as hydro, wind, and solar power [33]. Related to HCI, some researchers have focused on developing technologies for harvesting renewable energy to power wearable devices and interactive systems [2, 11, 22, 40, 42]. While most of these technologies convert the harvested energy into electricity for storage and later use, recent work in the robotics field has demonstrated the feasibility of alternative pneumatic-energy-generating systems that can be more efficient, concise, and environmentally friendly [26, 27, 36].

Despite the growing interest in pneumatic tangible interfaces in the human-computer interaction (HCI) field [4, 6, 7, 13–17, 20, 23, 24, 29, 35, 37–39, 41, 43–45], most current pneumatic interfaces are still powered by electronic pumps and valves, which limits their application in broader contexts. To address this limitation, this work investigates how to harvest various ambient renewable energy sources, convert them to compressed air, and store the compressed air for later actuating utilization in a pre-programmed way.

We hope this research can promote the development of sustainable pneumatic interfaces that can operate autonomously in an ambient environment and potentially be used in a wide range of applications. The core contributions of this work [18] are as follows:

- A three-phase ambient energy harnessing strategy and solution for the pneumatic interface:
 - **Harvesting:** non-electric pumps that can harvest a variety of ambient energy sources and leverage this energy to compress air;
 - **Storage:** storage units with distinct volume-pressure characteristics to store the compressed air for different demands;
 - **Utilization:** non-electric valves that are triggered by environmental factors with tunable thresholds to manage the compressed air for actuation autonomously;
- A corresponding design tool that:
 - guides users through the process of designing the system;
 - provides estimation of working results for quick iterations;
 - exports material list and fabrication instructions to facilitate prototyping.

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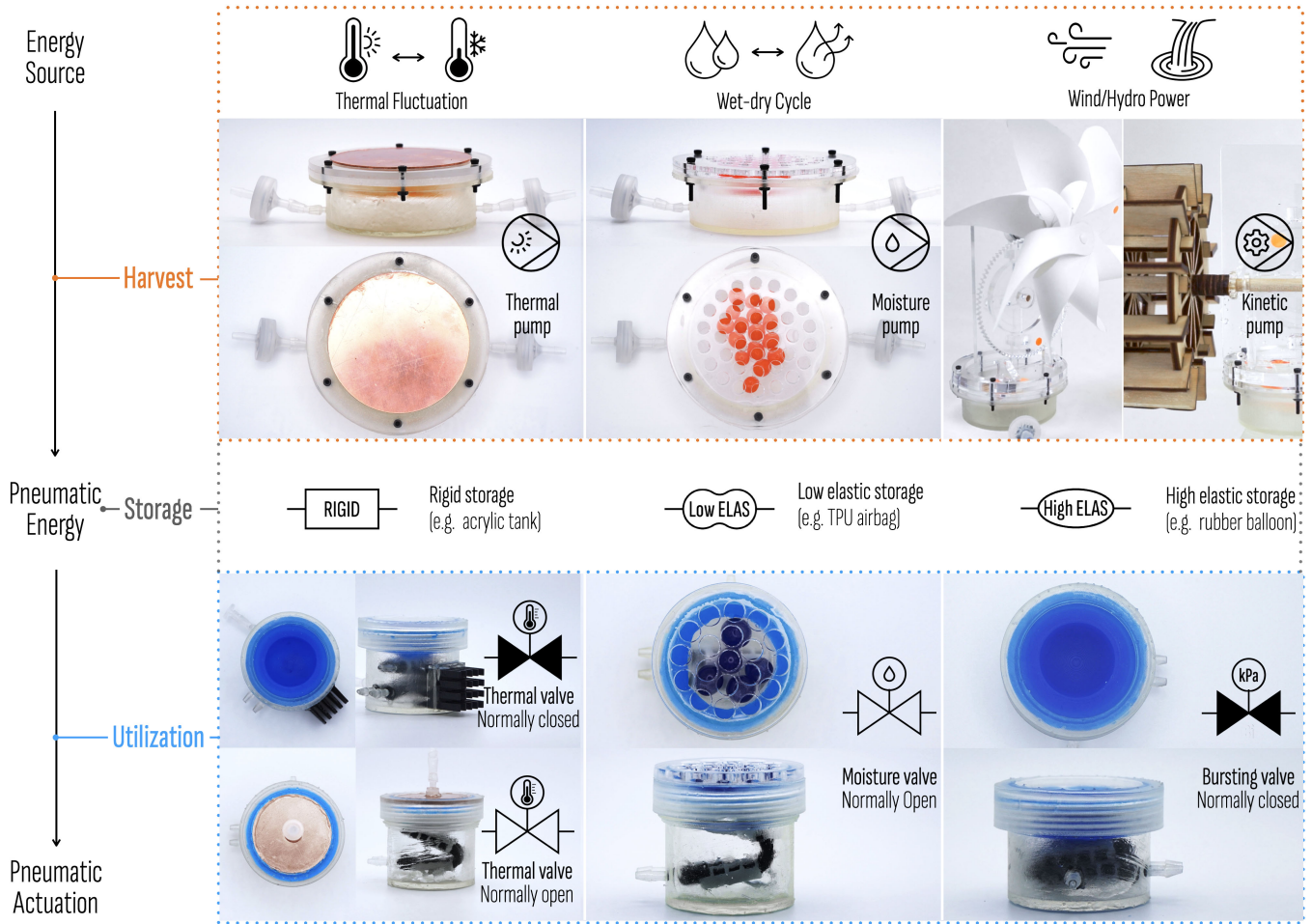


Figure 1: The three-phase energy harnessing solution provided by Sustainflatable.

- Functional performance evaluation of the Sustainflatable system.
- Several application examples that demonstrate the capability of the Sustainflatable system.

2 SUSTAINFLATABLE OVERVIEW

To harness ambient energy for a pneumatic interface, we propose a three-phase solution: harvesting, storing and utilizing.

2.1 Pumps for Energy Harvesting

2.1.1 Thermal Pump. A thermal pump compresses air by leveraging temperature fluctuations, with the bladder expanding to expel air from the outlet when the ambient temperature surpasses the boiling point of the LBL liquid and contracting to refill with air through the inlet when the temperature drops below the boiling point.

2.1.2 Moisture Pump. A moisture pump harnesses dry-wet cycles to compress air, with water beads swelling and expelling air from the outlet when the pump becomes wet, and deswelling to refill with air through the inlet when the pump dries out.

2.1.3 Kinetic Pump. A kinetic pump harnesses mechanical energy from wind power or hydro energy to compress air by converting rotational motion into linear reciprocating motion, where the downward movement of the slider pushes the membrane to expel compressed air through the outlet and vice versa.

2.2 Storage Units for Energy Accumulation

Storage units made of different materials will exhibit different volume and pressure characteristics. Rigid storage is suitable for rapidly building up the pressure but has limited volume efficiency. Highly elastic storage is ideal for high volume applications with slower pressure buildup. Low elastic storage offers a balanced choice for both metrics.

2.3 Stimuli Responsive Valves for Energy Utilization

The valves in this system rely on tube kinking[19] and/or bistable membrane phenomena for operation[1]. Tube kinking occurs when a flexible tube is bent or compressed, obstructing the flow path. The

bistable semi-spherical membrane can exist in two stable equilibrium states. Additionally, we have developed reconfigurable valves triggered by environmental factors, enhancing adaptability and autonomy.

2.3.1 Bursting Valve. A bursting valve is designed to respond to changes in air pressure and either open or close accordingly. The bistable membrane enables the valve to snap upward and un-kink the tubes when the control port's air pressure exceeds a specific threshold, keeping the valve open unless there is negative pressure in the control channel or external force on the membrane. A monostable membrane can reset and close the valve when air pressure decreases automatically.

2.3.2 Thermal Valve. A thermal valve responds to the ambient temperature changes with two types: normally closed (NC) and normally open (NO). Both types use LBL as the working medium, and the NC valve opens when the temperature exceeds the threshold, while the NO valve works oppositely. They are both reversible.

2.3.3 Moisture Valve. A moisture valve, activated by the dry-wet cycle, uses water beads as the working medium. When wet, the water beads swell, kinking the tubes by pushing the membrane, while they deswell and allowing the membrane to recover and the tubes to unkink when dry.

3 DESIGN TOOL

We developed a design tool to guide users through the design and fabrication process of Sustainflatable system. The design tool enables the users to explore different configurations and structures of the system with a simple GUI. It also provides a preview of the system's working procedure reacting to the environment changes. The preview function is implemented by a simplified idea gas model to analyze the states of the pneumatic components in the system with computational solutions. Lastly, the tool can export corresponding model, material list and assembly instructions to facilitate prototyping of the system. We developed the design tool on Rhino 7 Grasshopper platform.

3.1 User workflow

The overall user workflow of the design tool are introduced in the following four steps.

3.1.1 Step 1. Design a Sustainflatable system. First, the user designs a Sustainflatable system by configuring a few pneumatic components based on a pre-defined template, which comprises four connected elements in series: a pump, valve1, storage, and valve2. For each component, users can configure basic type and most of the related functional parameters (Fig. 2). The user can get a real-time structure diagram in three-dimensional preview window in the design tool.

3.1.2 Step 2. Configure the environmental cycle inputs. Next, the user configures the environmental condition with a series of *cycles*. Here, a *cycle* refers to a shortest period of environmental change that may lead to a periodic working processes of the system, which contains a round-change of temperature and a constant moisture and wind condition. Users can carefully design the condition of each cycle in the design tool (Fig. 2).

Step 1. Design

Select and configure the pump, storage, and valves

Pump Type:

Storage Type: Max volume (mL): Stable pressure (kPa):

Valve 1 Orientation: Type: Normal state: Triggered state: Open pressure (kPa): Close pressure (kPa): Trigger:

Valve 2 Orientation: Type: Normal state: Triggered state: Trigger temperature (°C):

Step 2. Environmental Cycles

Configure the cycle inputs

Cycle count:

Temperature (°C) start: middle: end:

Moisture/water bath: ☒

Windspeed (m/s):

CYCLE 3

CYCLE 4

CYCLE 5

CYCLE 6

CYCLE 7

CYCLE 8

Step 3. Preview

Preview the result and iterate

Update Preview

Cycle:

Temperature (°C):

Moisture/water bath: ☒

Windspeed (m/s):

Step 4. Export the design

Model & material list and instructions

Export model and material list

Figure 2: Design tool GUI that allows users to input design configurations.

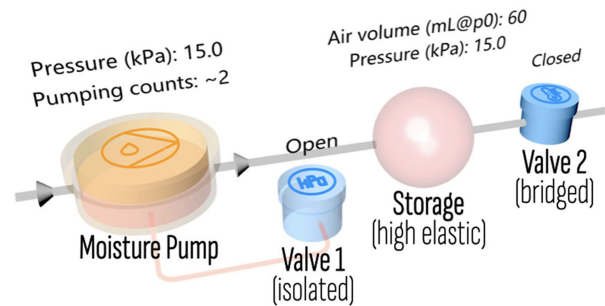


Figure 3: Visualization and simulation based on the design configuration selected in Fig. 2, where the volume changing of the components is visualized by three-dimensional models, the pressure changing by red shades, and the open/closed state of the valves by translucent/solid appearance.

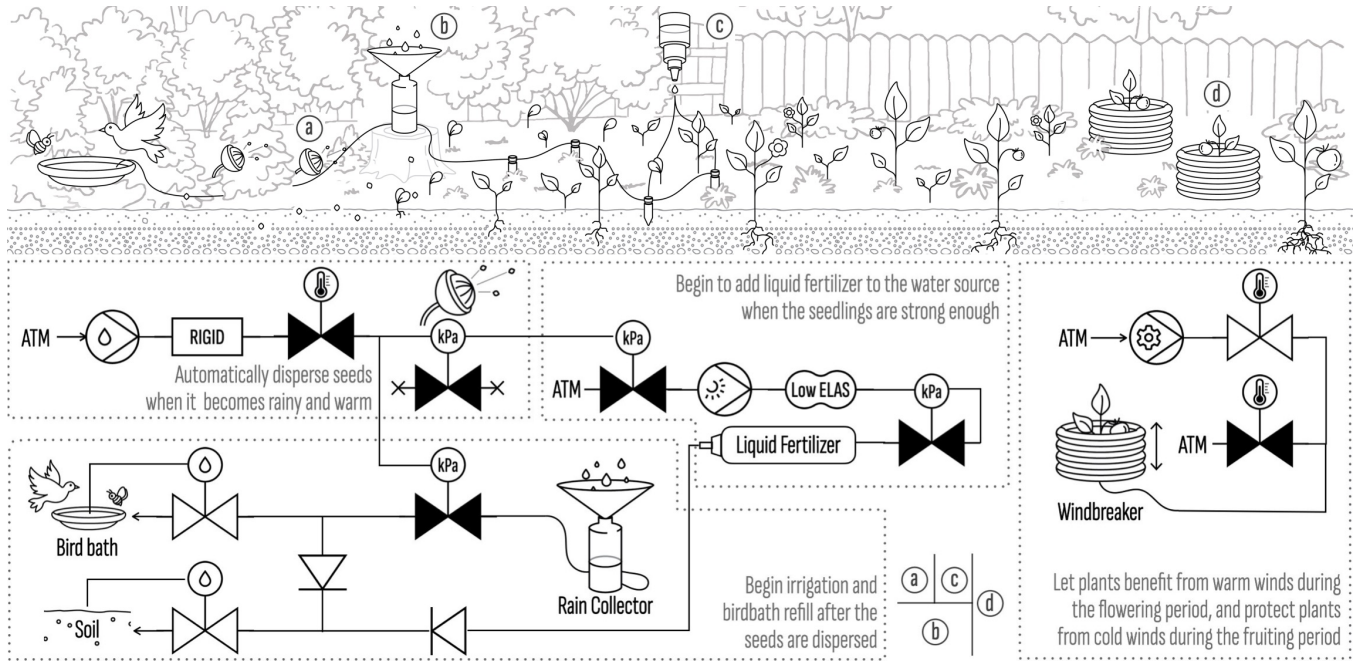


Figure 4: Overview of application examples. a). Automatic seeds ejection under the right time and conditions. b). Subsequent self-feedback irrigation and bird baths fill. c). Appropriately timed fertilization when the seedlings are strong enough. d). Self-regulating plant insulation for flowering and fruiting period.

3.1.3 Step 3. Generate preview results. Then, the user hits the 'Update Preview' button, slides the timeline to watch the preview of the system's working procedure corresponding to the environmental cycle inputs. The design tool provides previews of a few status parameters, e.g., the pressure and the pumping volume of the pump, the pressure and the air amount of the storage, and the states of the valves (Fig 3). These previews are expected to provide an intuitive understanding for the users before their actual application. Users can go back to step 2 to iterate the system design if the previewed function is not satisfying.

3.1.4 Step 4. Export the components list. When finishing the Sustainflatable system design, the user exports the corresponding model and material list which will facilitate the subsequent fabrication process of the system.

4 APPLICATION EXAMPLES

The Sustainflatable system is designed to be versatile and adaptable, making it suitable for deployment in various settings, whether indoors or outdoors. It is also autonomous, meaning that it does not require constant human intervention or monitoring. In this section, we present several examples of how the system can be used in a backyard garden context (Fig. 4). These concepts serve to illustrate the potential of our technique and inspire HCI researchers and designers to develop even more innovative and sophisticated applications in a wide range of scenarios.

4.1 Automatic Seeds Ejection under Suitable Conditions

Some plants have evolved unique and effective seed dispersal mechanisms. For example, when squirting cucumber's fruit is ripe, any slight touch or disturbance can cause it to burst open, ejecting the seeds inside with a forceful squirt¹.

Inspired by natural phenomena, we have developed an automatic seed dispersal system, illustrated in Fig. 4.a. Imagine a garden in a temperate continental climate zone, where winters are cold and dry but springs are characterized by frequent rain. In the system, a moisture pump will compress air towards the storage unit when spring starts, accumulating pressure. The successive thermal valve ensures that compressed air is only released to the bursting valve when it is warm enough. Once the necessary pressure has been accumulated and the temperature is favorable, the bursting valve ejects the seeds with an initial velocity of approximately 4 m/s. The bursting valve features a customized body and a customized membrane [12] that enables faster snap-through speed.

4.2 Self-feedback Irrigation and Bird Baths Fill

Timely irrigation plays a crucial role in successful seed germination. Moreover, placing bird baths in the garden can attract pollinators and predators of pests [31][25]. By creating a welcoming environment for these beneficial creatures, we are promoting a healthy and sustainable ecosystem in the garden.

As shown in Fig. 4.b, the rainwater collector is controlled by a bursting valve which will open simultaneously with the seeding

¹Wikipedia - Squirting Cucumber

dispersal. This will enable rainwater to flow into the irrigation and birdbath fill system. Successive moisture valves will halt the irrigation and refill processes once the soil is sufficiently moist and the birdbath is full, and will resume the processes when the soil dries out and the water level in the birdbath drops. Due to the relatively slow response time of the moisture valves, we have incorporated ceramic porous rods to prevent over-irrigation and over-refilling by allowing water to seep out gradually. In addition, to prevent contamination of the birdbath refill system by the fertilizer which will be added later, a check valve is installed in front of the mixing point.

4.3 Appropriately Timed Fertilization

Fertilizers can promote plant growth and ensure optimal yields. However, the timing of fertilizer application is crucial. For instance, applying fertilizer to a newly germinated seedling can lead to fertilizer burn, causing damage or even death [32].

The appropriately timed liquid fertilizer distribution system we developed is depicted in Fig. 4.c. At the same time as the seeding dispersal, the bursting valve opens, unblocking the inlet of the thermal pump and allowing it to generate compressed air based on daily temperature fluctuations. After a specified number of days, when the seedlings are strong enough, the second bursting valve bursts, allowing the compressed air to squeeze the liquid fertilizer out and mix it with rainwater before being applied to the plants via the irrigation system. To prevent the backflow of the irrigation rainwater, a check valve is installed in front of the mixing point. The waiting time before distributing the fertilizer can be estimated and optimized using our design tool ahead.

4.4 Self-regulating Plant Insulation

Wind is an important factor in the growth and survival of plants, providing several benefits such as promoting pollination [3], improving transpiration [9], and pest control [30]. However, a cold, strong wind in the late fall can cause excessive water loss through transpiration, leading to dehydration and wilting. Additionally, the harsh winds can cause delayed ripening or even chilling injury to the fruits [21].

To harness the advantages of wind while mitigating potential hazards, we have implemented a self-regulating insulation system, as demonstrated in Fig. 4.d. The wind pump can be positioned to face the prevailing wind direction during autumn. During warm weather before the fall, the NO thermal valve closes while the NC thermal valve opens. This prevents air from being pumped into the inflatable windbreaker, enabling the plants to benefit from the wind. When there is cold wind, the NO thermal valve opens while the NC thermal valve closes and the wind pump will operate, gradually inflating the windbreaker and shielding the plants from the wind. This automated process ensures that the windbreaker provides the necessary protection to the plants during inclement weather without many human interventions.

5 CONCLUSION

In conclusion, the Sustainflatable system offers a sustainable solution to power and control pneumatic interfaces. The system's ability to harvest renewable energy sources, convert the energy

into compressed air, and store it for later use in a programmable and intelligent way, make it a cost-effective, energy-efficient and environmentally friendly alternative to traditional electric pumps and valves. The system's electronic-free design, incorporating customized energy harvesting pumps, storage units with variable volume-pressure characteristics, and tailored valves that operate autonomously, demonstrates its ability to significantly improve the sustainability of pneumatic interfaces.

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REFERENCES

- [1] G. W. Brodland and H. Cohen. 1987. Deflection and snapping of spherical caps. *International Journal of Solids and Structures* 23, 10 (Jan. 1987), 1341–1356. [https://doi.org/10.1016/0020-7683\(87\)90001-1](https://doi.org/10.1016/0020-7683(87)90001-1)
- [2] Young-Man Choi, Moon Gu Lee, and Yongho Jeon. 2017. Wearable Biomechanical Energy Harvesting Technologies. *Energies* 10, 10 (Oct. 2017), 1483. <https://doi.org/10.3390/en10101483>
- [3] Theresa M. Culley, Stephen G. Weller, and Ann K. Sakai. 2002. The evolution of wind pollination in angiosperms. *Trends in Ecology & Evolution* 17, 8 (Aug. 2002), 361–369. [https://doi.org/10.1016/S0169-5347\(02\)02540-5](https://doi.org/10.1016/S0169-5347(02)02540-5)
- [4] Juri Fujii, Satoshi Nakamaru, and Yasuaki Kakehi. 2021. LayerPump: Rapid Prototyping of Functional 3D Objects with Built-in Electrohydrodynamics Pumps Based on Layered Plates. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '21)*. Association for Computing Machinery, New York, NY, USA, 1–7. <https://doi.org/10.1145/3430524.3442453>
- [5] Azmat Gani. 2021. Fossil fuel energy and environmental performance in an extended STIRPAT model. *Journal of Cleaner Production* 297 (May 2021), 126526. <https://doi.org/10.1016/j.jclepro.2021.126526>
- [6] Teng Han, Fraser Anderson, Pourang Irani, and Tovi Grossman. 2018. HydroRing: Supporting Mixed Reality Haptics Using Liquid Flow. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology (UIST '18)*. Association for Computing Machinery, New York, NY, USA, 913–925. <https://doi.org/10.1145/3242587.3242667>
- [7] Liang He, Gierad Laput, Eric Brockmeyer, and Jon E. Froehlich. 2017. SqueezePulse: Adding Interactive Input to Fabricated Objects Using Corrugated Tubes and Air Pulses. In *Proceedings of the Eleventh International Conference on Tangible, Embedded, and Embodied Interaction (TEI '17)*. Association for Computing Machinery, New York, NY, USA, 341–350. <https://doi.org/10.1145/3024969.3024976>
- [8] Richard Leslie Hills. 1996. *Power from Wind: A History of Windmill Technology*. Cambridge University Press, Cambridge, England. Google-Books-ID: FoVkf-BV1_8C.
- [9] Cheng-Wei Huang, Chia-Ren Chu, Cheng-I Hsieh, Sari Palmroth, and Gabriel G. Katul. 2015. Wind-induced leaf transpiration. *Advances in Water Resources* 86 (Dec. 2015), 240–255. <https://doi.org/10.1016/j.advwatres.2015.10.009>
- [10] Environmental and Energy Study Institute. 2020. Fossil Fuels | EESI. <https://www.eesi.org/topics/fossil-fuels/description>
- [11] Vikram Iyer, Hans Gaensbauer, Thomas L. Daniel, and Shyamnath Gollakota. 2022. Wind dispersal of battery-free wireless devices. *Nature* 603, 7901 (March 2022), 427–433. <https://doi.org/10.1038/s41586-021-04363-9>
- [12] Lishuai Jin, Yueying Yang, Bryan O. Torres Maldonado, Sebastian David Lee, Nadia Figueroa, Robert J. Full, and Shu Yang. 2023. Ultrafast, Programmable, and Electronics-Free Soft Robots Enabled by Snapping Metacaps. *Advanced Intelligent Systems* n/a, n/a (Feb. 2023), 2300039. <https://doi.org/10.1002/aisy.202300039>
- [13] Jasmine Lu, Ziwei Liu, Jas Brooks, and Pedro Lopes. 2021. Chemical Haptics: Rendering Haptic Sensations via Topical Stimulants. In *The 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21)*. Association for Computing Machinery, New York, NY, USA, 239–257. <https://doi.org/10.1145/3472749.3474747>
- [14] Qiuyu Lu, Chengpeng Mao, Liyuan Wang, and Haipeng Mi. 2016. LIME: Liquid MEtal Interfaces for Non-Rigid Interaction. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (Tokyo, Japan) (UIST '16)*. Association for Computing Machinery, New York, NY, USA, 449–452. <https://doi.org/10.1145/2984511.2984562>
- [15] Qiuyu Lu, Jifei Ou, João Wilbert, André Haben, Haipeng Mi, and Hiroshi Ishii. 2019. milliMorph – Fluid-Driven Thin Film Shape-Change Materials for Interaction Design. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology (UIST '19)*. Association for Computing Machinery, New York, NY, USA, 663–672. <https://doi.org/10.1145/3332165.3347956>

- [16] Qiuyu Lu, Danqing Shi, Yingqing Xu, and Haipeng Mi. 2020. MetalLife: Interactive Installation Based on Liquid Metal Deformable Interfaces. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3334480.3383134>
- [17] Qiuyu Lu, Haiqing Xu, Yijie Guo, Joey Yu Wang, and Lining Yao. 2023. Fluidic Computation Kit: Towards Electronic-free Shape-changing Interfaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23). Association for Computing Machinery, New York, NY, USA, 1–21. <https://doi.org/10.1145/3544548.3580783>
- [18] Qiuyu Lu, Tianyu Yu, Semina Yi, Yuran Ding, Haipeng Mi, and Lining Yao. 2023. SustainInflate: Harvesting, Storing and Utilizing Ambient Energy for Pneumatic Morphing Interfaces. In *Proceedings of the 36th Annual Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, 1–20. <https://doi.org/10.1145/3586183.3606721>
- [19] Kai Luo, Philipp Rothmund, George M. Whitesides, and Zhigang Suo. 2019. Soft kink valves. *Journal of the Mechanics and Physics of Solids* 131 (Oct. 2019), 230–239. <https://doi.org/10.1016/j.jmps.2019.07.008>
- [20] Yiyue Luo, Kui Wu, Andrew Spielberg, Michael Foshey, Daniela Rus, Tomás Palacios, and Wojciech Matusik. 2022. Digital Fabrication of Pneumatic Actuators with Integrated Sensing by Machine Knitting. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (CHI '22). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3491102.3517577>
- [21] James M. Lyons. 1973. Chilling Injury in Plants. *Annual Review of Plant Physiology* 24, 1 (1973), 445–466. <https://doi.org/10.1146/annurev.pp.24.060173.002305>
- [22] Yogesh Kumar Meena, Krishna Seunarine, Deepak Ranjan Sahoo, Simon Robinson, Jennifer Pearson, Chi Zhang, Matt Carnie, Adam Pockett, Andrew Prescott, Suzanne K. Thomas, Harrison Ka Hin Lee, and Matt Jones. 2020. PV-Tiles: Towards Closely-Coupled Photovoltaic and Digital Materials for Useful, Beautiful and Sustainable Interactive Surfaces. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3313831.3376368>
- [23] Haipeng Mi, Meng Wang, Qiuyu Lu, and Yingqing Xu. 2018. Tangible user interface: origins, development, and future trends. *SCIENTIA SINICA Informationis* 48, 4 (2018), 390–405. <https://doi.org/10.1360/N112017-00227>
- [24] Takafumi Morita, Yu Kuwajima, Ayato Minaminosono, Shingo Maeda, and Yasuaki Kakehi. 2022. HydroMod: Constructive Modules for Prototyping Hydraulic Physical Interfaces. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (CHI '22). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3491102.3502096>
- [25] PennState Department of Agricultural Sciences. 2023. Pollinator Garden Certification. <https://ento.psu.edu/research/centers/pollinators/public-outreach/cert>
- [26] Kazunori Ogawa, Tomohiro Ikeda, and Yuichi Kurita. 2018. Unplugged Powered Suit for Superhuman Tennis. In *2018 12th France-Japan and 10th Europe-Asia Congress on Mechatronics*. IEEE Press, New York, NY, USA, 361–364. <https://doi.org/10.1109/MECATRONICS.2018.8495845>
- [27] Kazunori Ogawa, Chetan Thakur, Tomohiro Ikeda, Toshio Tsuji, and Yuichi Kurita. 2017. Development of a pneumatic artificial muscle driven by low pressure and its application to the unplugged powered suit. *Advanced Robotics* 31, 21 (Nov. 2017), 1135–1143. <https://doi.org/10.1080/01691864.2017.1392345>
- [28] A. G. Olabi and Mohammad Ali Abdelkareem. 2022. Renewable energy and climate change. *Renewable and Sustainable Energy Reviews* 158 (April 2022), 112111. <https://doi.org/10.1016/j.rser.2022.112111>
- [29] Jifei Ou, Mélina Skouras, Nikolaos Vlavianos, Felix Heibeck, Chin-Yi Cheng, Jannik Peters, and Hiroshi Ishii. 2016. aeroMorph - Heat-sealing Inflatable Shape-change Materials for Interaction Design. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* (UIST '16). Association for Computing Machinery, New York, NY, USA, 121–132. <https://doi.org/10.1145/2984511.2984520>
- [30] Judith E. Pasek. 1988. 30. Influence of wind and windbreaks on local dispersal of insects. *Agriculture, Ecosystems & Environment* 22-23 (Aug. 1988), 539–554. [https://doi.org/10.1016/0167-8809\(88\)90044-8](https://doi.org/10.1016/0167-8809(88)90044-8)
- [31] Marissa V. Ponder, Gordon W. Frankie, Rachel Elkins, Kate Frey, Rollin Coville, Mary Schindler, Sara Leon Guerrero, Jaime C. Pawelek, and Carolyn Shaffer. 2013. How to Attract and Maintain Pollinators in Your Garden. *n/a n/a, n/a* (Oct. 2013), 13. <https://doi.org/10.3733/ucanr.8498>
- [32] Drew Swainston published. 2023. When to fertilize seedlings – to encourage strong and healthy growth. <https://www.homesandgardens.com/gardens/when-to-fertilize-seedlings>
- [33] Atika Qazi, Fayaz Hussain, Nasrudin ABD. Rahim, Glenn Hardaker, Daniyal Alghazzawi, Khaled Shaban, and Khalid Haruna. 2019. Towards Sustainable Energy: A Systematic Review of Renewable Energy Sources, Technologies, and Public Opinions. *IEEE Access* 7 (2019), 63837–63851. <https://doi.org/10.1109/ACCESS.2019.2906402>
- [34] Terry S. Reynolds. 1983. *Stronger Than a Hundred Men: A History of the Vertical Water Wheel*. JHU Press, n/a. Google-Books-ID: aZ0runvvrq0AC.
- [35] Harpreet Sareen, Udayan Umapathi, Patrick Shin, Yasuaki Kakehi, Jifei Ou, Hiroshi Ishii, and Pattie Maes. 2017. PrintInflate: Printing Human-Scale, Functional and Dynamic Inflatable Objects. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (CHI '17). Association for Computing Machinery, New York, NY, USA, 3669–3680. <https://doi.org/10.1145/3025453.3025898>
- [36] Rachel A. Shveda, Anoop Rajappan, Te Faye Yap, Zhen Liu, Marquise D. Bell, Barclay Jumeat, Vanessa Sanchez, and Daniel J. Preston. 2022. A wearable textile-based pneumatic energy harvesting system for assistive robotics. *Science Advances* 8, 34 (Aug. 2022), eabo2418. <https://doi.org/10.1126/sciadv.abo2418>
- [37] Saya Suzunaga, Yuichi Itoh, Yuki Inoue, Kazuyuki Fujita, and Takao Onoye. 2020. TuVe: A Shape-changeable Display using Fluids in a Tube. In *Proceedings of the International Conference on Advanced Visual Interfaces* (AVI '20). Association for Computing Machinery, New York, NY, USA, 1–9. <https://doi.org/10.1145/3399715.3399874>
- [38] Carlos E. Tejada, Raf Ramakers, Sebastian Boring, and Daniel Ashbrook. 2020. AirTouch: 3D-printed Touch-Sensitive Objects Using Pneumatic Sensing. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3313831.3376136>
- [39] Shan-Yuan Teng, Tzu-Sheng Kuo, Chi Wang, Chi-huan Chiang, Da-Yuan Huang, Liwei Chan, and Bing-Yu Chen. 2018. PuPoP: Pop-up Prop on Palm for Virtual Reality. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology* (UIST '18). Association for Computing Machinery, New York, NY, USA, 5–17. <https://doi.org/10.1145/3242587.3242628>
- [40] Shan-Yuan Teng, K. D. Wu, Jacqueline Chen, and Pedro Lopes. 2022. Prolonging VR Haptic Experiences by Harvesting Kinetic Energy from the User. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (UIST '22). Association for Computing Machinery, New York, NY, USA, 1–18. <https://doi.org/10.1145/3526113.3545635>
- [41] Marynel Vázquez, Eric Brockmeyer, Ruta Desai, Chris Harrison, and Scott E. Hudson. 2015. 3D Printing Pneumatic Device Controls with Variable Activation Force Capabilities. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (CHI '15). Association for Computing Machinery, New York, NY, USA, 1295–1304. <https://doi.org/10.1145/2702123.2702569>
- [42] Chen Xu, Yu Song, Mengdi Han, and Haixia Zhang. 2021. Portable and wearable self-powered systems based on emerging energy harvesting technology. *Microsystems & Nanoengineering* 7, 1 (March 2021), 1–14. <https://doi.org/10.1038/s41378-021-00248-z>
- [43] Junichi Yamaoka, Ryuma Niiyama, and Yasuaki Kakehi. 2017. BlowFab: Rapid Prototyping for Rigid and Reusable Objects using Inflation of Laser-cut Surfaces. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology* (UIST '17). Association for Computing Machinery, New York, NY, USA, 461–469. <https://doi.org/10.1145/3126594.3126624>
- [44] Junichi Yamaoka, Kazunori Nozawa, Shion Asada, Ryuma Niiyama, Yoshihiro Kawahara, and Yasuaki Kakehi. 2018. AccordionFab: Fabricating Inflatable 3D Objects by Laser Cutting and Welding Multi-Layered Sheets. In *Adjunct Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology* (UIST '18 Adjunct). Association for Computing Machinery, New York, NY, USA, 160–162. <https://doi.org/10.1145/3266037.3271636>
- [45] Lining Yao, Ryuma Niiyama, Jifei Ou, Sean Follmer, Clark Della Silva, and Hiroshi Ishii. 2013. PneuUI: pneumatically actuated soft composite materials for shape changing interfaces. In *Proceedings of the 26th annual ACM symposium on User interface software and technology* (UIST '13). Association for Computing Machinery, New York, NY, USA, 13–22. <https://doi.org/10.1145/2501988.2502037>